

Hammad Ahmed

API 571 – Corrosion and Material | API 580 – Risk-Based Inspection | CENOSCO IMS PEI (v1) | Pakistan Engineering Council Accreditation | Metallurgical & Materials Engineer

Year of Birth: 2001

Nationality: Pakistani

Address: New Bhogiwal Baghbanpura, Lahore Passport No: -----

Email: hammadahmed1111996@gmail.com

Mobile Number: 0092 316 4168156

Skype: -----

LinkedIn: www.linkedin.com/in/hammad-ahmed-aa24b6244

Qualification / Educational Background:

- B.Sc Metallurgy & Materials Engineering
(Institute of Metallurgy & Material Engineering University of the Punjab, Lahore), (2019-2023)
- Pakistan Engineering Council Accreditation

Professional Trainings / Certifications

- API 571 – Corrosion and Material (Cert No.129029)
- API 580 – Risk-Based Inspection (Cert No.125983)
- CENOSCO IMS PEI (v1)
- AMPP – Basics of Cathodic Protection

Publications:

- Hammad Ahmed et al 2025 Eng. Res. Express 7 035435. Effect of tool rotational speed on the mechanical and electrochemical properties of friction stir welded 5083 Al-Mg alloy.

Link: <https://iopscience.iop.org/article/10.1088/2631-8695/ae04c6>

EXPERIENCE

SGS Pakistan Pvt. Ltd. (2024 to Present):



Corrosion Engineer

(Corrosion Engineer-Pakistan: Pipeline, Cathodic Protection, Internal Corrosion, Coatings)
(Corrosion Engineer-Middle East Region: Pipeline. Cathodic Protection, Internal Corrosion, Coatings)

Summary of Experience:

Over the past +1 years, after completing my degree in Metallurgy & Materials Engineering, I have successfully contributed to more than 20+ projects related to Onshore Pipeline Integrity Management, Internal Corrosion Projects and Cathodic Protection Systems, including EPC and turnkey solutions covering design, procurement, installation, and commissioning. My experience also includes CP audits, troubleshooting, system rehabilitation, coating inspections, internal corrosion solutions, and corrosion monitoring assessments using coupons, probes, and data loggers. These projects have been executed for clients across various sectors such as Oil & Gas, Power, Fertilizers, and Chemicals within Pakistan.

I'm responsible to handle projects assignments as:

- Pipeline Integrity Management (Oil and Gas Up-Stream, Mid-Stream & Down-Stream Pipelines),
- Cathodic Protection Systems EPC - Turnkey Solutions – (Design, Procurement, Installation & Commissioning), CP Auditing Troubleshooting Rehabilitations & Consultancy,
- Develop complex technical CP designs including calculations, material specification, QA/QC documents & engineering drawings
- Integrity Assessments particularly NACE ECDA-External Corrosion Direct Assessment Methodologies, NACE ICDA-Internal Corrosion Direct Assessments Methodologies & NACE SCCDA-Stress Corrosion Cracking Direct Assessment Methodologies,
- Manage site survey teams for CIPS, DCVG, ACVG, ACCA, PCM, Soil Resistivity Surveys, Pipeline Elevation Profile Survey, CP Audits, Underground Pipeline/Cables Locating & Mapping, pH Evaluation (in the ditch examination), Interference /Stray Current Surveys

- Bell-Holes Management (in the ditch evaluation) NDT Visual Inspections ASME B31G Assessments & decisions for Run-Repair-Replace-Rerate & Pipe/Coating/CP repair reinstate QAQC & backfilling, auditing, trainings, Projects Management.
-

Major Projects:

- As a **Project Engineer** responsible for the Rehabilitation of existing CP System of whole plant, Supply of material, Installation & Testing of **Balloki Power Plant**. (2024 to 2025)
 - As a **Project Engineer** responsible for the CP EPC (transformer rectifier based) ICCP system - detailed engineering design, field surveys, supply, installation, connection, testing, pre-commissioning, and follow-up testing of complete and fully functional impressed current cathodic protection system ICCP, of complete underground pipeline of HRS extension at **PSO terminal Allama Iqbal International Airport Lahore**. (2025 to 2025)
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Other Training / Course - During Engineering:

2022 (Internship/Training) Descon Technical Institute, Lahore



API INDIVIDUAL CERTIFICATION PROGRAMS



verifies that

Hammad Ahmed

HAS MET THE ESTABLISHED AND PUBLISHED REQUIREMENTS FOR API CERTIFICATION AS AN

API 571 CORROSION AND MATERIALS INSPECTOR

CERTIFICATION NUMBER **129029**

ORIGINAL CERTIFICATION DATE **August 31, 2025**

CURRENT CERTIFICATION DATE **August 31, 2025**

EXPIRATION DATE **August 31, 2028**

DIRECTOR, API INDIVIDUAL CERTIFICATION PROGRAMS



ICP



To verify authenticity, scan the above QR code
or go to <http://inspectorsearch.api.org>

API INDIVIDUAL CERTIFICATION PROGRAMS



verifies that

Hammad Ahmed

HAS MET THE ESTABLISHED AND PUBLISHED REQUIREMENTS FOR API CERTIFICATION AS AN

API 580 RISK-BASED INSPECTION INSPECTOR

CERTIFICATION NUMBER **125983**

ORIGINAL CERTIFICATION DATE **April 30, 2025**

CURRENT CERTIFICATION DATE **April 30, 2025**

EXPIRATION DATE **April 30, 2028**

DIRECTOR, API INDIVIDUAL CERTIFICATION PROGRAMS



To verify authenticity, scan the above QR code
or go to <http://inspectorsearch.api.org>.

Certificate

This certificate acknowledges that

Hammad Ahmed

has successfully completed the course

IMS PEI: Complete (v1)

February 8, 2025

Credit Hours: 32

Fundamentals, Circuits, Degradation Management, IMS Features

Luiz Curado
Rowan Vaduganathan

Luiz Curado
Rowan Vaduganathan



CERTIFICATE OF PROFESSIONAL DEVELOPMENT HOURS

Be it known

Hammad Ahmed

has attended an AMPP sponsored meeting on

Basics of Cathodic Protection

In recognition whereof, **1.00** Professional Development Hour(s) are awarded.

16 July 2024

DATE

Tasos Kostrivas

Dr. Tasos Kostrivas

CHAIR - AMPP UK CHAPTER

رجسٹریشن نمبر
Registration Number
2019-ucc-69

University of the Punjab



جامعہ پنجاب

رول نمبر
Roll Number
194141E-S1-330

This university hereby confers upon
Hammad Ahmed

son of **Khalid Mehmood**

of the **Institute of Metallurgy and Materials Engineering,
University of the Punjab, Lahore**

یہ جامعہ

حماد احمد

بن **خالد محمود**

از انسٹیٹیوٹ آف میٹالرجی اینڈ میٹیریئلز انجینئرنگ
جامعہ پنجاب، لاہور

the degree of

**Bachelor of Science (Engineering)
Metallurgy & Materials Engineering**

together with all the rights, privileges and
honours belonging to this degree.

Session 2019-2023

C.G.P.A. 3.64

کو بی۔ ایس سی (انجینئرنگ)

میٹالرجی اینڈ میٹیریئلز انجینئرنگ

کی سند عطا کرتی ہے مع تمام حقوق، امتیازات

اور اعزازات جو اس سند کے ساتھ وابستہ ہیں۔

دورانیہ ۲۰۱۹-۲۰۲۳ء

سی۔ جی۔ بی۔ ای۔ ۳.۶۴



محمد بلیمغ الرحمن

Chancellor

چانسلر

Controller of Examinations

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Lahore : Pakistan

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Effect of tool rotational speed on the mechanical and electrochemical properties of friction stir welded 5083 Al-Mg alloy

Hammad Ahmed¹ , Muhammad Arslan Hafeez² , Muhammad Bilal Asghar¹ , Usama Sarwar¹ , Muhammad Haris Qureshi¹ , Muhammad Afzal³ and Ameerq Farooq^{1,*}

¹ Corrosion Control Research Cell, Institute of Metallurgy and Materials Engineering, Faculty of Chemical and Materials Engineering, University of the Punjab, Lahore, 54590, Pakistan

² School of Materials Science and Engineering, State Key Laboratory of Materials Processing and Die & Mould Technology, Huazhong University of Science and Technology, 430074 Wuhan, Hubei, People's Republic of China

³ National Institute of Vacuum Science & Technology (NINVEST) NCP Complex, Quaid-e-Azam University, Islamabad-3125, Pakistan

* Author to whom any correspondence should be addressed.

E-mail: ameeq.farooq@gmail.com

Keywords: friction stir welding, aluminum-magnesium alloys, phase analysis, mechanical analysis, corrosion analysis

Abstract

While friction stir welding (FSW) is a sustainable method for joining aluminum alloys, the effect of rotational speed on the combined microstructural, mechanical, and corrosion behavior of 5083 Al-Mg alloy is not well established. Most studies treat these properties separately, creating a gap in understanding their interdependence. Closing this gap is essential for optimizing FSW parameters for durable marine applications. Therefore, this study investigates the impact of varying rotational speeds (600, 700, and 800 RPM) on the metallurgical evolution, mechanical performance, and electrochemical behavior of friction stir welded 5083 Al-Mg alloy. Welds were produced using a butt joint configuration with a fixed tool traverse speed of 180 mm min⁻¹. Comprehensive analyses were conducted, encompassing x-ray diffraction analysis, mechanical testing, and electrochemical corrosion evaluations. The findings reveal that a rotational speed of 800 RPM delivers the most optimal weld characteristics compared to 600 RPM and 700 RPM. Specifically, the weld fabricated at this speed exhibited the highest tensile strength (342.48 MPa), minimal residual stresses, a highly uniform phase formation, and exceptional corrosion resistance, evidenced by an impressively low corrosion rate of 0.01 mpy.

1. Introduction

Aluminum alloys are highly valued for their exceptional attributes, including an impressive strength-to-weight ratio, outstanding corrosion resistance, and cost-efficiency [1, 2]. Among these, the 5083 aluminum alloy distinguishes itself with superior performance in extreme environments, demonstrating exceptional resistance to seawater and industrial chemicals [3]. Moreover, this alloy retains its mechanical strength even after welding, making it ideal for use in challenging and demanding applications [4]. Primarily alloyed with magnesium and supplemented by minor amounts of manganese, chromium, and silicon, 5083 Al-Mg alloy is a non-heat-treatable, strain-hardened alloy widely used in marine, automotive, aerospace, and structural applications due to its excellent corrosion resistance, high strength-to-weight ratio, and good weldability. In spite of these diverse applications, there is a discrepancy in joining this alloy perfectly [5]. This is because conventional fusion welding often causes defects such as porosity, hot cracking, and loss of mechanical properties in this alloy. This fact highlights the pressing need to develop advanced and optimized joining techniques to maximize its potential and utility [6].

Friction stir welding (FSW), being a solid-state joining technique, avoids melting-related problems and produces defect-free joints with refined microstructures and improved corrosion resistance [7, 8]. This method is particularly well-suited for the 5083 Al-Mg alloy, offering significant advantages over conventional fusion